Spark Streaming Program (Scala)

```
C:\Users\91988>spark-shell
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 4.1.1
      /_/

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 17.0.18)
Type in expressions to have them evaluated.
Type :help for more information.
Spark context Web UI available at http://LAPTOP-6DEICVVQ:4040
Spark context available as 'sc' (master = local[*], app id = local-1777009385296).
Spark session available as 'spark'.

scala> import spark.implicits._
import spark.implicits._

scala> spark.sparkContext.setLogLevel("ERROR")

scala> val lines = spark.readStream.format("socket").option("host", "localhost").option("port", 9999).load()
val lines: org.apache.spark.sql.DataFrame = [value: string]

scala> val words = lines.as[String]
val words: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val stopWords = Seq("is", "the", "a", "an", "and", "to")
val stopWords: Seq[String] = List(is, the, a, an, and, to)
```

```
scala> val cleaned = words.map { line =>
     |   val lower = line.toLowerCase
     |   val noSpace = lower.trim.replaceAll("\\s+", " ")
     |   val noPunct = noSpace.replaceAll("[^a-zA-Z ]", "")
     |   val filtered = noPunct.split(" ").filter(word => !stopWords.contains(word))
     |   val lemmatized = filtered.map {
     |     case w if w.endsWith("ing") => w.dropRight(3)
     |     case w if w.endsWith("ed")  => w.dropRight(2)
     |     case w => w
     |   }
     |   lemmatized.mkString(" ")
     | }
val cleaned: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val query = cleaned.writeStream.outputMode("append").format("console").option("checkpointLocation", "C:/tmp/check
point").start()
val query: org.apache.spark.sql.streaming.StreamingQuery = org.apache.spark.sql.execution.streaming.runtime.StreamingQueryWrapper@53f6a24

-------------------------------------------
Batch: 0
-------------------------------------------
+-----+
|value|
+-----+
+-----+

-------------------------------------------
Batch: 1
-------------------------------------------
+------------------+
|             value|
+------------------+
|she danc sing song|
+------------------+
```

```
-------------------------------------------
Batch: 2
-------------------------------------------
+---------------------+
|                value|
+---------------------+
|cats are play in ...|
+---------------------+

-------------------------------------------
Batch: 3
-------------------------------------------
+---------------------+
|                value|
+---------------------+
|he walk talk teacher|
+---------------------+
```

```
Windows PowerShell
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PS C:\Users\91988> $listener = [System.Net.Sockets.TcpListener]::new([System.Net.IPAddress]::Any, 9999)
PS C:\Users\91988> $listener.Start()
PS C:\Users\91988> Write-Host "Waiting for Spark to connect on port 9999..."
Waiting for Spark to connect on port 9999...
PS C:\Users\91988> $client = $listener.AcceptTcpClient()
PS C:\Users\91988> Write-Host "Connected! Type text and press Enter:"
Connected! Type text and press Enter:
PS C:\Users\91988> $writer = [System.IO.StreamWriter]::new($client.GetStream())
PS C:\Users\91988> $writer.AutoFlush = $true
PS C:\Users\91988> while($true) {
>>     $input = Read-Host
>>     $writer.WriteLine($input)
>> }The dog is running and jumping
She is dancing and singing a song
The cats are playing in the garden
He walked and talked to the teacher
```